Stage 1: Mathematical Formulation Notes

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1 Stage 1: Mathematical Formulation Notes

1.1 Purpose

Record the American option variational inequality and implementation notation before coding.

1.2 Method

Use time-to-maturity $\tau = T - t$ and normalize experiments to $K = 1$ unless otherwise stated.

1.3 Mathematical Formulation

$$dS_t = (r - q)S_t dt + \sigma S_t dW_t, \quad \mathcal{L}U = \frac{1}{2}\sigma^2 S^2 U_{SS} + (r - q)SU_S - rU.$$

$$U \geq \Phi, \quad U_\tau - \mathcal{L}U \geq 0, \quad (U - \Phi)(U_\tau - \mathcal{L}U) = 0.$$

The continuation premium is $C = U - \Phi \geq 0$.

1.4 Implementation Details

Inputs use $(\log(S/K), \tau/T, r, q, \sigma, T, 1_{\text{call}})$ for surrogate datasets.

1.5 Validation and Diagnostics

Check operator sign, payoff, boundary conventions, and Greek units before implementation.

1.6 Results

This stage produces notation and sign conventions only.

1.7 Failure Modes and Limitations

The main risks are mixing calendar time and time-to-maturity or reversing the LCP sign.

1.8 Conclusion and Next Step

Proceed to CN/PSOR implementation and validation.